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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (Honours)
V SEMESTER
Discipline Specific Core Course - MG5DSCMAT300
A FIRST COURSE IN COMPLEX ANALYSIS
2024 ADMISSION ONWARDS

Duration: 2 hours **Maximum Marks:** 70

Remember(K), Understand (U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Answer any 5 questions *Each question carries 2 marks*

1. State the Extended Cauchy Integral Formula for the nth derivative of an analytic function. **[U] / [CO3]**

   - Ans: If $f(z)$ is analytic inside and on a simple closed contour C (positively oriented), and z_0 is any point interior to C, then $f(z)$ has derivatives of all orders at z_0, given by:
   $$f^{(n)}(z_0) = \frac{n!}{2\pi i}\int_C \frac{f(z)}{(z-z_0)^{n+1}}\,dz \quad (n = 0, 1, 2, \dots)$$

   2 marks

2. What is the parametrization of line segment joining z_1 and z_2? **[K] / [CO1]**

   - Ans:
   $$z(t) = (1-t)z_1 + tz_2, \quad 0 \le t \le 1$$
   or $z(t) = z_1 + t(z_2 - z_1)$ for $t \in [0, 1]$. 2 marks

3. Find the Maclaurin series for $f(z) = z\sin z$. **[U] / [CO6]**

   - Ans:
   $$\sin z = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+1} = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots \quad (|z| < \infty)$$

   1 mark

   $$f(z) = z\sin z = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+2} = z^2 - \frac{z^4}{3!} + \frac{z^6}{5!} - \dots \quad (|z| < \infty)$$

   2 marks

4. State Laurent's theorem. **[K] / [CO6]**

- Ans: Let f be analytic throughout an annular domain $D : R_1 < |z - z_0| < R_2$ centered at z_0, and let C denote any positively oriented simple closed contour around z_0 lying in D. Then at each point z in D, $f(z)$ can be represented by the series:

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n}$$

2 marks

where:

$$a_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}}\, dz \quad (n = 0, 1, 2, \dots)$$

$$b_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{-n+1}}\, dz \quad (n = 1, 2, 3, \dots)$$

5. What is the integral representation of the residue of a function f? **[K] / [CO3, CO5]**

- Ans: If $f(z)$ is analytic in a deleted neighborhood of an isolated singular point z_0, and C is a positively oriented simple closed contour surrounding z_0 containing no other singular points, the residue of $f(z)$ at z_0 is defined as:

$$\text{Res}(f, z_0) = \frac{1}{2\pi i} \int_C f(z)\, dz$$

2 marks

6. What are the singular points, if any, of the function $\frac{\sin z^2}{z^2+1}$? **[K] / [CO2]**

- Ans: $z = i$ 1 mark
  and $z = -i$. 1 mark

7. If $z = e^{i\theta}$, express $\cos\theta$ in terms of z. **[U] / [CO7]**

- Ans: Using Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = z^{-1}$:

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{z + z^{-1}}{2} = \frac{z^2 + 1}{2z}$$

2 marks

8. Write $\sin 2\theta$ in terms of $z = e^{i\theta}$. **[E] / [CO7]**

- Ans: Since $z = e^{i\theta}$, we have $z^2 = e^{i2\theta}$ and $z^{-2} = e^{-i2\theta}$.

$$\sin(2\theta) = \frac{e^{i2\theta} - e^{-i2\theta}}{2i} = \frac{z^2 - z^{-2}}{2i} = \frac{z^4 - 1}{2iz^2}$$

2 marks

$[2 \times 5 = 10]$

Part B

Answer any 5 questions *Each question carries 6 marks*

9. Define a simply connected domain. Also, evaluate $\int_C \frac{ze^z}{(z^2+9)^5}dz$, where C denotes any closed contour lying in the open disk $|z| < 1$. **[U] / [CO2]**

   - Ans: A domain D in the complex plane is called **simply connected** if every simple closed contour C lying entirely within D encloses only points that belong to D.
   - 2 marks
   - The integrand $f(z) = \frac{ze^z}{(z^2+9)^5}$ has singular points where $z^2 + 9 = 0 \implies z = \pm 3i$.
   - The distance of both singular points from the origin is $|\pm 3i| = 3$.
   - Since the contour C lies inside $|z| < 1$, both singular points $z = \pm 3i$ lie **outside** the contour C.
   - Thus, $f(z)$ is analytic on and inside C. 2 marks
   - By Cauchy-Goursat Theorem:

$$\int_C \frac{ze^z}{(z^2+9)^5}dz = 0$$

   2 marks

10. Show that using an antiderivative for every contour extending from a point z_1 to a point z_2, $\int_C z^n dz = \frac{1}{n+1}(z_2^{n+1} - z_1^{n+1})$, $(n = 0, 1, 2, \dots)$. **[A] / [CO3]**

    - Ans: Let $f(z) = z^n$ for $n = 0, 1, 2, \dots$. Since $f(z)$ is an entire function, it possesses a continuous antiderivative:

$$F(z) = \frac{z^{n+1}}{n+1}$$

    where $F'(z) = f(z) = z^n$. 3 marks

    By the Fundamental Theorem of Complex Integration, if $f(z)$ has a continuous antiderivative $F(z)$ in a domain containing a contour C extending from z_1 to z_2:

$$\int_C z^n dz = [F(z)]_{z_1}^{z_2} = F(z_2) - F(z_1) = \frac{1}{n+1}(z_2^{n+1} - z_1^{n+1})$$

    3 marks

11. Assume that the complex series $\sum z_n$ is convergent. Show that $\lim_{n\to\infty} z_n = 0$. **[U] / [CO2]**

    - Ans: Let $S_N = \sum_{k=1}^{N} z_k$ be the sequence of partial sums of the series $\sum_{n=1}^{\infty} z_n$. Given that $\sum z_n$ converges to a limit S, we have:

$$\lim_{N\to\infty} S_N = S$$

    2 marks

    For $N \geq 2$, the N-th term can be written as $z_N = S_N - S_{N-1}$. 1 mark

    Taking the limit as $N \to \infty$:

$$\lim_{N\to\infty} z_N = \lim_{N\to\infty}(S_N - S_{N-1}) = \lim_{N\to\infty} S_N - \lim_{N\to\infty} S_{N-1} = S - S = 0$$

    3 marks

    Hence, $\lim_{n\to\infty} z_n = 0$.

12. (a) Define Taylor series. **[C] / [CO6]**

- Ans: Let $f(z)$ be analytic throughout an open disk $|z - z_0| < R$. Then $f(z)$ can be expanded in a **Taylor series** about z_0:

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z - z_0)^n$$

2 marks

(b) Show that the Maclaurin series representation of $\frac{1}{1-z}$ is $\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$ $(|z| < 1)$. **[C] / [CO6]**

- Ans: When $z_0 = 0$, the Taylor series is called a **Maclaurin series**. Let $f(z) = \frac{1}{1-z} = (1-z)^{-1}$. Taking derivatives at $z_0 = 0$:
  - $f(0) = 1 = 0!$
  - $f'(z) = (1-z)^{-2} \implies f'(0) = 1 = 1!$
  - $f''(z) = 2(1-z)^{-3} \implies f''(0) = 2 = 2!$ 2 marks
  - $f^{(n)}(z) = n!(1-z)^{-(n+1)} \implies f^{(n)}(0) = n!$

  Substituting into the Maclaurin series formula:

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} z^n = \sum_{n=0}^{\infty} \frac{n!}{n!} z^n = \sum_{n=0}^{\infty} z^n \quad (|z| < 1)$$

2 marks

13. Show that the singular point of the following functions is a pole. Determine the order of that pole and the corresponding residue. **[U] / [CO2]**

(a) (a) $\frac{1-\cosh z}{z^3}$

(b) (b) $\frac{1-e^{2z}}{z^4}$

- Ans:

(a) $f(z) = \frac{1-\cosh z}{z^3}$

Using the Taylor expansion of $\cosh z = 1 + \frac{z^2}{2!} + \frac{z^4}{4!} + \ldots$:

$$1 - \cosh z = -\left(\frac{z^2}{2!} + \frac{z^4}{4!} + \frac{z^6}{6!} + \ldots\right)$$

Divide by z^3:

$$f(z) = \frac{1}{z^3}\left(-\frac{z^2}{2!} - \frac{z^4}{4!} - \ldots\right) = -\frac{1}{2!\,z} - \frac{z}{4!} - \frac{z^3}{6!} - \ldots$$

- The principal part has a single term $\frac{-1/2}{z}$.
- Therefore, $z = 0$ is a **pole of order 1** (simple pole).
- The residue is the coefficient of $\frac{1}{z}$, so $\text{Res}(f, 0) = -\frac{1}{2}$. 3 marks

(b) $f(z) = \frac{1-e^{2z}}{z^4}$

Using the Taylor expansion of $e^{2z} = 1 + 2z + \frac{(2z)^2}{2!} + \frac{(2z)^3}{3!} + \ldots$:

$$1 - e^{2z} = -2z - 2z^2 - \frac{4}{3}z^3 - \frac{2}{3}z^4 - \ldots$$

Divide by z^4:

$$f(z) = -\frac{2}{z^3} - \frac{2}{z^2} - \frac{4/3}{z} - \frac{2}{3} - \ldots$$

- The highest power of $\frac{1}{z}$ in the principal part is 3, so $z = 0$ is a **pole of order 3**.
- The residue is the coefficient of $\frac{1}{z}$, so $\text{Res}(f, 0) = -\frac{4}{3}$. 3 marks

14. Determine the singularities of $f(z) = \frac{z^2}{(z-1)^2(z+2)}$ and evaluate the corresponding residues. **[U] / [CO2]**

- Ans: The denominator is zero at $z = 1$ and $z = -2$. 2 marks

(a) **At $z = -2$:** It is a **pole of order 1** (simple pole).

$$\text{Res}(f, -2) = \lim_{z\to -2}(z+2)f(z) = \lim_{z\to -2}\frac{z^2}{(z-1)^2} = \frac{(-2)^2}{(-2-1)^2} = \frac{4}{9}$$

2 marks

(b) **At $z = 1$:** It is a **pole of order 2**.

$$\text{Res}(f, 1) = \lim_{z\to 1}\frac{d}{dz}\left[(z-1)^2 f(z)\right] = \lim_{z\to 1}\frac{d}{dz}\left(\frac{z^2}{z+2}\right)$$

Using quotient rule:

$$\frac{d}{dz}\left(\frac{z^2}{z+2}\right) = \frac{2z(z+2) - z^2(1)}{(z+2)^2} = \frac{z^2+4z}{(z+2)^2}$$

Evaluating at $z = 1$:

$$\text{Res}(f, 1) = \frac{1^2 + 4(1)}{(1+2)^2} = \frac{5}{9}$$

2 marks

15. Find the principal value of $\int_{-\infty}^{\infty} \frac{x}{x^2+9} dx$. **[E] / [CO7]**

- Ans: The function $g(x) = \frac{x}{x^2+9}$ is an **odd function** since $g(-x) = -g(x)$. By the definition of Cauchy Principal Value:

$$\text{P.V.}of \int_{-\infty}^{\infty} \frac{x}{x^2+9}\, dx = \lim_{R\to\infty}\int_{-R}^{R} \frac{x}{x^2+9}\, dx$$

3 marks

Since $g(x)$ is odd and integrated over a symmetric interval $[-R, R]$:

$$\int_{-R}^{R} \frac{x}{x^2+9}\, dx = 0 \quad \text{for all } R > 0 \implies \text{P.V.}\int_{-\infty}^{\infty} \frac{x}{x^2+9}\, dx = 0$$

3 marks

16. Find $\int_0^\infty \frac{dx}{x^4+16}$. **[E] / [CO7]**

- Ans: Since the integrand is even, $I = \int_0^\infty \frac{dx}{x^4+16} = \frac{1}{2}\int_{-\infty}^\infty \frac{dx}{x^4+16}$. 1 mark

  Consider $f(z) = \frac{1}{z^4+16}$ over a semicircular contour C_R in the upper half-plane. Poles occur at $z^4 = -16 = 16e^{i\pi}$, giving upper half-plane poles:

$$z_0 = 2e^{i\pi/4} = \sqrt{2} + i\sqrt{2}, \quad z_1 = 2e^{i3\pi/4} = -\sqrt{2} + i\sqrt{2}$$

  2 marks

  Residue at a simple pole z_k:

$$\text{Res}(f, z_k) = \frac{1}{4z_k^3} = \frac{z_k}{4z_k^4} = -\frac{z_k}{64}$$

  1 mark

  Sum of residues in upper half-plane:

$$\sum \text{Res} = -\frac{1}{64}(z_0 + z_1) = -\frac{1}{64}(2i\sqrt{2}) = -\frac{i\sqrt{2}}{32}$$

  By Residue Theorem:

$$\int_{-\infty}^\infty \frac{dx}{x^4+16} = 2\pi i\left(-\frac{i\sqrt{2}}{32}\right) = \frac{\pi\sqrt{2}}{16} \implies \int_0^\infty \frac{dx}{x^4+16} = \frac{\pi\sqrt{2}}{32}$$

  2 marks

[$6 \times 5 = 30$]

Part C

Answer any 3 questions *Each question carries 10 marks*

17. State and prove the Maximum Modulus Principle. **[An] / [CO1, CO4]**

- Ans: **Statement:** If a function f is analytic and not constant in a given domain D, then $|f(z)|$ has no maximum value in D. 2 marks

  Proof:

  – Since R is closed and bounded and $|f(z)|$ is continuous on R, $|f(z)|$ attains a maximum M at some point $z_0 \in R$.

  – Suppose z_0 is an interior point of R. Then there exists a disk $|z - z_0| \leq \rho$ lying entirely inside R.

  – By Gauss's Mean Value Theorem:

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + \rho e^{i\theta})\, d\theta$$

  4 marks

  – Taking absolute values:

$$|f(z_0)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + \rho e^{i\theta})|\, d\theta \leq \frac{1}{2\pi} \int_0^{2\pi} M\, d\theta = M$$

  – Since $|f(z_0)| = M$, we have:

$$\frac{1}{2\pi} \int_0^{2\pi} [M - |f(z_0 + \rho e^{i\theta})|]\, d\theta = 0$$

  – Since $M - |f(z_0 + \rho e^{i\theta})| \geq 0$ and is continuous, $|f(z)| = M$ for all points on $|z - z_0| = \rho$.

  – Since this holds for all $r \leq \rho$, $|f(z)| = M$ throughout the disk. An analytic function with constant modulus is constant.

  – Thus, if $f(z)$ is non-constant, $|f(z)|$ cannot attain its maximum in the interior and must occur on the boundary. 4 marks

  – **7 Marks are to be awarded for the proof according to the discretion of the evaluator**.

18. Prove that if $z_n = x_n + iy_n$ $(n = 1, 2, \dots)$ and $z = x + iy$, then $\lim_{n\to\infty} z_n = z$ if and only if $\lim_{n\to\infty} x_n = x$ and $\lim_{n\to\infty} y_n = y$. **[U] / [CO2]**

- Ans: ($\implies$) Suppose $\lim_{n\to\infty} z_n = z$. For every $\epsilon > 0$, $\exists N \in \mathbb{N}$ such that for all $n > N$, $|z_n - z| < \epsilon$. Using the inequalities $|x_n - x| \leq |z_n - z|$ and $|y_n - y| \leq |z_n - z|$:

$$|x_n - x| < \epsilon \quad \text{and} \quad |y_n - y| < \epsilon \quad \text{for all } n > N$$

  Thus, $\lim_{n\to\infty} x_n = x$ and $\lim_{n\to\infty} y_n = y$. 5 marks

  ($\impliedby$) Suppose $\lim_{n\to\infty} x_n = x$ and $\lim_{n\to\infty} y_n = y$. Given $\epsilon > 0$, $\exists N_1, N_2$ such that $|x_n - x| < \frac{\epsilon}{2}$ for $n > N_1$ and $|y_n - y| < \frac{\epsilon}{2}$ for $n > N_2$. Let $N = \max(N_1, N_2)$. By triangle inequality, for $n > N$:

$$|z_n - z| = |(x_n - x) + i(y_n - y)| \leq |x_n - x| + |y_n - y| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

  Hence, $\lim_{n\to\infty} z_n = z$. 5 marks

19. Expand $f(z) = \frac{-1}{(z-1)(z-2)}$ in powers of z and state the region of validity. **[A] / [CO6]**

- Ans: By partial fractions:

$$f(z) = \frac{-1}{(z-1)(z-2)} = \frac{1}{z-1} - \frac{1}{z-2}$$

1 mark

- **Region 1 ($|z| < 1$):**

$$f(z) = -\frac{1}{1-z} + \frac{1}{2(1-z/2)} = \sum_{n=0}^{\infty} \left(\frac{1}{2^{n+1}} - 1 \right) z^n$$

3 marks

- **Region 2 ($1 < |z| < 2$):**

$$f(z) = \frac{1}{z(1-1/z)} + \frac{1}{2(1-z/2)} = \sum_{n=0}^{\infty} \frac{1}{z^{n+1}} + \sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}$$

3 marks

- **Region 3 ($|z| > 2$):**

$$f(z) = \frac{1}{z(1-1/z)} - \frac{1}{z(1-2/z)} = \sum_{n=0}^{\infty} \frac{1-2^n}{z^{n+1}}$$

3 marks

20. Let $f(z) = \frac{z^2}{z^2-1}$.

(a) Find all the poles of $f(z)$ and determine their order.

- Ans: $f(z) = \frac{z^2}{(z-1)(z+1)}$. 2 marks
  - The denominator is zero at $z = 1$ and $z = -1$. 2 marks
  - Since numerator $z^2 \neq 0$ at these points, both $z = 1$ and $z = -1$ are simple poles. 2 marks

(b) Find the residue of $f(z)$ at each pole. **[A] / [CO5]**

- Ans
  -

$$\text{Res}(f, 1) = \lim_{z \to 1} (z-1) \frac{z^2}{(z-1)(z+1)} = \frac{1^2}{1+1} = \frac{1}{2}$$

2 marks

$$\text{Res}(f, -1) = \lim_{z \to -1} (z+1) \frac{z^2}{(z-1)(z+1)} = \frac{(-1)^2}{-1-1} = -\frac{1}{2}$$

2 marks

21. Using residues, evaluate the improper integral $\int_{-\infty}^{\infty} \frac{x \sin ax}{x^2+4} dx \quad (a > 0)$. **[E] / [CO7]**

- Ans: Consider $f(z) = \frac{ze^{iaz}}{z^2+4}$. The upper half-plane pole is $z = 2i$ (simple pole). 2 marks

$$\text{Res}(f, 2i) = \lim_{z \to 2i} (z-2i) \frac{ze^{iaz}}{(z-2i)(z+2i)} = \frac{2ie^{-2a}}{4i} = \frac{e^{-2a}}{2}$$

2 marks

By Residue Theorem:

$$\int_{-\infty}^{\infty} \frac{xe^{iax}}{x^2+4}\,dx = 2\pi i\left(\frac{e^{-2a}}{2}\right) = i\pi e^{-2a}$$

2 marks

Taking imaginary parts:

$$\int_{-\infty}^{\infty} \frac{x\sin(ax)}{x^2+4}\,dx = \pi e^{-2a}$$

4 marks

Marks are to be awarded for the steps according to the discretion of the evaluator.

22. **[A] / [CO7]**

(a) State and explain Jordan's Lemma.

- Ans: (Jordan's Lemma) - Let $f(z)$ be analytic at all points in the upper half plane $y \geq 0$ that are exterior to a circle $|z| = R_0$, $C_R : z = Re^{i\theta}$ $(0 \leq \theta \leq \pi)$, where $R > R_0$ and for all points z on C_R, there is a positive constant M_R such that $|f(z)| \leq M_R$ and $\lim_{R\to\infty} M_R = 0$, then for any $a > 0$:

$$\lim_{R\to\infty} \int_{C_R} f(z)e^{iaz}\,dz = 0$$

3 marks

Explanation can be given upto 2 marks. Total for part (a) is 3 marks.

(b) Find the Cauchy principal value of the integral $\int_{-\infty}^{\infty} \frac{x\sin x}{x^2+2x+2}dx$.

- Ans: Consider $f(z) = \frac{ze^{iz}}{z^2+2z+2}$. The upper half-plane pole is $z_0 = -1+i$.

2 marks

$$\text{Res}(f, -1+i) = \frac{(-1+i)e^{i(-1+i)}}{2i} = \frac{e^{-1}}{2}\left[(\cos 1 + \sin 1) + i(\cos 1 - \sin 1)\right]$$

By Residue Theorem:

$$\int_{-\infty}^{\infty} \frac{xe^{ix}}{x^2+2x+2}dx = 2\pi i\,\text{Res}(f, -1+i) = \pi e^{-1}\left[-(\cos 1 - \sin 1) + i(\cos 1 + \sin 1)\right]$$

2 marks

Taking imaginary part:

$$\text{P.V.}\int_{-\infty}^{\infty} \frac{x\sin x}{x^2+2x+2}dx = \frac{\pi}{e}(\cos 1 + \sin 1)$$

3 marks

$[10 \times 3 = 30]$

NB. Marks are to be awarded proportionally for other right methods used for getting the answers or part of the answers.